Plan of Action

2026 RSNA Knee Abnormality Detection AI Challenge

https://www.kaggle.com/competitions/rsna-knee-abnormality-detection

Team: Group 2

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1. Data Collection and Exploratory Data Analysis

Before building any model, we will first get to know the data well, so that our later choices are based on what we actually see in it.

What we will look at

- **train.csv:** the twelve binary abnormality labels and the multilingual *Report* field.
- **train_series.csv:** for each MRI series, its anatomical plane, fluid sensitivity, and fat suppression.

What we will explore (EDA)

- Number of slices per series
- How MRI orientations are distributed
- How often each label occurs, and which labels tend to occur together
- Which languages the reports are written in, and how long they are

What we expect to find

This analysis should show us any **missing labels**, differences in **image quality**, and **class imbalance**. These findings will guide our preprocessing and model design.

2. Data Preprocessing and Label Preparation

Images

- Standardize the DICOM intensity scaling and normalize the pixel intensities.
- Resize every slice to 256 × 256 pixels.
- Group the MRI series by anatomical plane (Axial, Coronal, Sagittal), fluid sensitivity, and fat suppression.
- Build 2.5D or 3D volumetric inputs, taking care to keep the important anatomical information.

Text and extra supervision

- Tokenize the multilingual radiology reports using a pretrained transformer such as XLM-RoBERTa.
- Extract abnormality-related information from the reports.
- Where a study has missing labels, use reliable, high-confidence labels taken from its report as extra (auxiliary) supervision.

Avoiding data leakage

- Keep every series and slice from the same StudyInstanceUID in the same data split, so no study appears in both training and validation.

3. AI Model Development and Training

Model design

- A 3D Convolutional Neural Network (CNN) or a Vision Transformer will extract features from the MRI scans across the different anatomical planes.
- A lightweight multilingual text encoder will process the radiology reports.

Our main idea: text-guided visual attention (cross-attention fusion)

We will build a cross-attention module in which the report's token embeddings help guide where the model looks in the spatial MRI feature maps during training. This lets the model learn from both the scans and the diagnostic text, which carry complementary information.

Reports are **not provided for the test studies**, so the model must work without them. To prepare for this:

- We apply text-modality dropout during training.
- We add a learnable null token (or zero-masking) to stand in when text is unavailable.
- At inference time the model makes its predictions through an **image-only pathway**.

Training setup

- **Loss:** weighted multi-label Binary Cross-Entropy (BCE) to deal with class imbalance.
- **Report-derived labels:** trained with a confidence-weighted loss.
- **Memory:** mixed-precision training and gradient accumulation to cope with the GPU memory needed for 3D MRI inputs.

4. Model Validation and Performance Optimisation

Validation

- Group-stratified K-Fold cross-validation using StudyInstanceUID, keeping abnormality-label balance where possible and preventing study-level data leakage.

How we measure performance

- **Primary metric:** ROC-AUC for each abnormality, and the macro-averaged AUC across all twelve findings, which is the competition's primary evaluation metric.
- **Secondary metric:** weighted log loss, to track the quality of our predicted probabilities.

How we improve it

- Use anatomically appropriate Test-Time Augmentation (TTA), average predictions across folds, and ensemble models where it helps.
- Optimize using validation AUC rather than classification thresholds, because the main challenge metric does not depend on a threshold. Threshold tuning will be used only for extra, supplementary evaluation.

Final model and predictions

- Choose the best-performing configuration based on the cross-validation results.
- Produce confidence scores for all twelve abnormalities using **MRI images alone** during final inference.